ASSIGNMENT - 6

① $A^{\mu} = \gamma_u^{\nu}\left[\gamma_u^{\nu}\,\frac{\vec{u}\cdot\vec{a}}{c}\,\vec{a} + \left(\frac{\vec{u}\cdot\vec{a}}{c^2}\,\vec{u}\right)\gamma_u^{\nu}\right]$

$$|A| = A^{\mu}\cdot A_{\mu} = |A_0|^2 - \vec{A}\cdot\vec{A}$$

$$= \gamma_u^4\left[\left(\gamma_u^2\,\frac{\vec{u}\cdot\vec{a}}{c^2}\right)^2 - \left(\vec{a} + \gamma_u^2\,\frac{\vec{u}\cdot\vec{a}}{c^2}\vec{u}\right)^2\right]$$

Recall that : $(\vec{x}+\vec{y})^2 = \vec{x}\cdot\vec{x} + 2\vec{x}\cdot\vec{y} + \vec{y}\cdot\vec{y}$

$$= x^2 + 2\vec{x}\cdot\vec{y} + y^2$$

[x & y are the norms of $\vec{x}$ & $\vec{y}$]

$$|A| = \left[\gamma_u^4\,\frac{(\vec{u}\cdot\vec{a})^2}{c^2} - a^2 - 2\gamma_u^2\,\frac{(\vec{u}\cdot\vec{a})^2}{c^2} - \gamma_u^4\,\frac{\vec{u}\cdot\vec{a}}{c^4}\,u^2\right]\gamma_u^4$$

$$= \gamma_u^4\left[\gamma_u^4\,\frac{(\vec{u}\cdot\vec{a})^2}{c^2}\underbrace{\left(1-\frac{u^2}{c^2}\right)}_{\frac{1}{\gamma_u^2}} - a^2 - 2\gamma_u^2\,\frac{(\vec{u}\cdot\vec{a})^2}{c^2}\right]$$

$$= \gamma_u^4\left[\gamma_u^2\,\frac{(\vec{u}\cdot\vec{a})^2}{c^2} - a^2 - 2\gamma_u^2\,\frac{(\vec{u}\cdot\vec{a})^2}{c^2}\right]$$

$$= -\gamma_u^4\left[a^2 + \gamma_u^2\,\frac{(\vec{u}\cdot\vec{a})^2}{c^2}\right]$$

In ICF ; $A'^{\mu} = (0, \vec{a}_0)$

$$|A'| = -\vec{a}_0\cdot\vec{a}_0 = -a_0^2$$

The norm of 4 vectors ; i.e., $X^{\mu}X_{\mu}$ is invariant (under both rotat^n & L.T.)

$$\therefore a_0^2 = \gamma_u^6\left[a^2\cdot\frac{1}{\gamma_u^2} + \frac{(u\,a\cos\theta)^2}{c^2}\right]$$

$$= \gamma_u^6\left[a^2(1-\beta_u^2) + \beta_u^2 a^2\cos^2\theta\right]$$

$$= \gamma_u^6\left[1-\beta_u^2 + \beta_u^2\cos^2\theta\right]a^2$$

$$= \gamma_u^6\left[1-\beta_u^2\sin^2\theta\right]a^2$$

$$\Rightarrow \boxed{a_0 = \gamma_u^3\left[1-\beta_u^2\sin^2\theta\right]^{1/2}a}$$

(2) $A_\mu = \begin{bmatrix} 5 \\ 1 \\ -2 \\ 0 \end{bmatrix}$ L is the Lorentz Transformation matrix

$A'^\mu = L A^\mu$

$A'_\mu = (L^T)^{-1} A_\mu$

$$L = \frac{1}{64}\begin{bmatrix} 125 & -48 & -60 & 75 \\ -75 & 80 & 36 & -45 \\ -60 & 0 & 80 & -36 \\ 48 & 0 & 0 & 80 \end{bmatrix}$$

$$(L^T)^{-1} = \frac{1}{64}\begin{bmatrix} 125 & 48 & 60 & -75 \\ 75 & 80 & 36 & -45 \\ 60 & 0 & 80 & -36 \\ -48 & 0 & 0 & 80 \end{bmatrix}$$

$$A'_\mu = \frac{1}{64}\begin{bmatrix} 125 & 48 & 60 & -75 \\ 75 & 80 & 36 & -45 \\ 60 & 0 & 80 & -36 \\ -48 & 0 & 0 & 80 \end{bmatrix}\begin{bmatrix} 5 \\ 1 \\ -2 \\ 0 \end{bmatrix}$$

$$= \frac{1}{64}\begin{bmatrix} 553 \\ 383 \\ 140 \\ -240 \end{bmatrix}$$

(3)(a) $L^T \eta L = \eta$

L_1 & L_2 ~~both~~ both are LT matrix

$L_1^T \eta L_1 = \eta$ — (I) $\qquad L_2^T \eta L_2 = \eta$ — (II)

$L_2^T \eta L_2 = L_2^T L_1^T \eta L_1 L_2 = (L_1 L_2)^T \eta (L_1 L_2)$

$L_2^T \eta L_2 = \eta \qquad \therefore (L_1 L_2)^T \eta (L_1 L_2) = \eta$

$\therefore L_1 L_2$ is also a LT matrix (proved)

(b) $|L_1| = +1$; $|L_2| = +1$

$|L_1 L_2| = |L_1||L_2| = 1$ (proved)

(c) L_1 & L_2 are 2 LT, carried out in succession.

L_1 has 0-0 components $\Lambda^0_0 \geq 1$

L_2 " " " $\Lambda^0_0 \geq 1$

So, we can say by topological argument that if we construct L_1 & L_2 successively, then L_1 & L_2 must have 0-0 components $\Lambda^0_0 \geq 1$

L_1 has 0-0 component $\Lambda^0_0 \geq 1$ by changing L_1 matrix by small amount we can perform L_2 from L_1 as L_2 also has $\Lambda^0_0 \geq 1$.

So, $L_1 L_2$ has $\Lambda^0_0 \geq 1$.